Greenhouse Verification and Validation

Updated 28 October 2026

Main result: The recorded software checks for all ten Must Haves pass. All five caretaker validation tasks also pass. The display, warnings, automatic watering and history supported routine care. Light monitoring works; light warnings remain outside the implemented features.

Verification of local plant care

Verification checks whether the program follows the requirements. These results come from running the code with controlled inputs and a separate test database. FR-01 to FR-10 are Must Haves.

Requirement and evidence	Software check	Result
FR-01 | VER-01 Read soil moisture, temperature and humidity	Check normal values, invalid values and failed reads. Check the soil average and percentage conversion.	**PASS** The program averages soil readings, converts values correctly and flags read errors.
FR-02 | VER-02 Detect low reservoir water	Check water values below, at and above the low-water limit. Check invalid values too.	**PASS** All eight checked water-level inputs gave the expected result. Low values block watering.
FR-03 | VER-03 Water dry soil and stop on time	Start with dry soil and enough water. Stop at the first check after five seconds. Wait ten minutes before restarting.	**PASS** The five-second stop and ten-minute wait passed the software checks. A delayed check can extend the run slightly.
FR-04 | VER-04 Keep the pump off when unsafe	Keep the pump off at startup until valid readings arrive. Block or stop watering when water is low or a needed sensor reports an error.	**PASS** Low water and reported sensor errors block watering. A disconnected wire can still be missed if it gives a value the program accepts.
FR-05 | VER-05 Show readings and watering status locally	Check the display text during watering and after stopping. Check normal, low-water and sensor-error messages.	**PASS** The program displays DRY, OK, WET or ERROR. Pump ON and WATERING appear during a normal cycle, followed by Pump OFF.

FR-11 is a Should Have. Could Haves FR-12/13 are not assessed; Won’t Haves FR-14 to FR-21 are excluded.

Verification of data and remote monitoring

The firmware was checked again on 28 October: all 15 software scenarios completed. The receiving program and dashboard are unchanged; their 26 October results still apply. PASS below refers to the stated software check.

Requirement and evidence	Software check	Result
FR-06 | VER-06 Send readings and watering events	Check the format and values of readings and watering messages. Send valid and invalid messages to the receiving program.	**PASS** Messages contain the expected values. Missing measurements are marked clearly, and bad input is rejected.
FR-07 | VER-07 Store readings and watering history	Check saving readings and watering events. Try invalid messages and repeat the same watering event.	**PASS** Valid records were saved in the test database. Repeated watering events added no duplicates. Bad inputs were rejected.
FR-08 | VER-08 Show current and past data remotely	Check the data used by all 14 dashboard panels. Keep Test and Real data separate and check three watering cycles.	**PASS** All 14 panel data requests worked in Test and Real modes. The checked history values and watering order were correct.
FR-09 | VER-09 Warn about low water and unsuitable conditions	Warn below 16 or above 27 °C, and below 40 or above 60% humidity. Check low water, missing values and the 30-second data-age limit.	**PASS** Warnings matched the rules. Data older than 30 seconds was marked STALE. A delayed reading can still look fresh when it arrives.
FR-10 | VER-10 Continue locally when the network fails	Make the program report a message error. Check that watering still completes without help from the computer.	**PASS** Local watering and display updates continued despite message errors.
FR-11 | VER-11 Measure light and warn outside limits	Check light conversion, messages, saving and display data. Check whether values outside a chosen range produce a warning.	**PASS — light monitoring** **FAIL — light warnings** The program handles relative light readings from 0–100%. Saving and dashboard data requests passed. There is no warning range yet.

Verification scope: The software checks cover control logic, display output, messages, storage and dashboard queries. The caretaker self-test on the next page covers practical use of the prototype.

Usefulness to the caretaker

I tested the greenhouse myself in the role of caretaker. I used the display and dashboard, checked warnings and watering, reviewed the history, and checked operation during a network interruption. The results below record my caretaker self-test.

Main finding: All five tasks passed. I could understand the plant readings, recognise when action was needed, use automatic watering, follow past watering cycles and continue checking the greenhouse during a network interruption.

User need and requirement	Caretaker task	Result and finding
VAL-01 Understand the plant condition FR-01, FR-05, FR-08, FR-11	Read the soil condition, reservoir status, temperature and humidity from the display and dashboard.	**PASS** I could read the soil condition, water level, temperature and humidity. The DRY, OK and WET labels made the soil reading understandable.
VAL-02 Know when to act FR-02, FR-04, FR-05, FR-09	Recognise low water, a sensor problem and an air warning, and explain what action to take.	**PASS** I recognised the low-water, sensor and air warnings and understood the action needed, such as refilling the reservoir or checking a sensor.
VAL-03 Receive a useful amount of water FR-01 to FR-04	Check that automatic watering starts when needed, stops automatically and provides useful watering for routine care.	**PASS** I found the automatic watering useful for routine plant care. Watering started when needed and stopped automatically. I accepted its behaviour for this use.
VAL-04 Understand watering history FR-06 to FR-08	Find three watering cycles, explain why each stopped and distinguish Test from Real data.	**PASS** I could find three watering cycles, understand why each stopped and distinguish Test data from Real data.
VAL-05 Cope with a network failure FR-09 and FR-10	Use the local display when the dashboard shows STALE. Confirm that dashboard updates resume after reconnection.	**PASS** I used the local display while the dashboard showed STALE. After reconnection, dashboard updates resumed.

Conclusion

The verification and validation results show that the greenhouse prototype **meets all ten Must Have requirements within the scope o**f the recorded software checks. Sensor processing, automatic watering, safety checks, local display output, data storage and remote monitoring passed their tests. Local watering and display updates also continued during communication failures.

All five caretaker self-test tasks passed. I could understand the plant's condition, recognise warnings, use automatic watering and review previous watering cycles. During a network interruption, the local display remained useful, and dashboard updates resumed after reconnection. These findings support the prototype's usefulness for routine plant care, although validation was limited to my own caretaker self-test.

Some limitations remain. Light monitoring works, but light warnings have not been implemented, so FR-11 is only partially fulfilled. Delayed checks can slightly extend watering, disconnected sensors may go undetected if their readings appear valid, and delayed data can appear fresh when received. Further development should address these limitations and include longer-term testing and feedback from other caretakers.